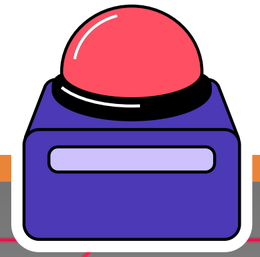
SGU INFORMATION TECHNOLOGY

INTRODUCTION TO OBJECT-ORIENTED PROGRAMMING

ALGORITHMS & OBJECT-ORIENTED PROGRAMMING
WEEK 10 LAB

JOSEPH ANDREAS



AGENDA

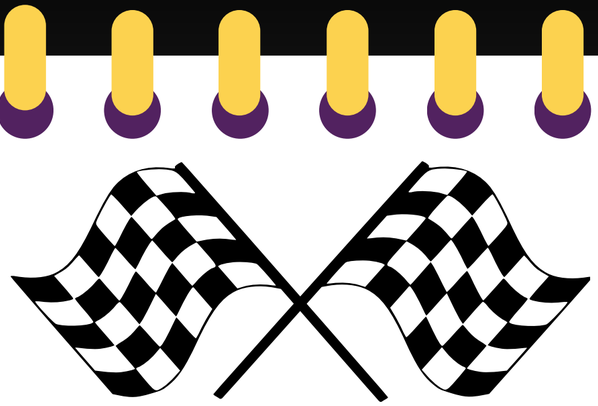
1. OBJECT-ORIENTED PROGRAMMING

2. CLASSES AND OBJECTS

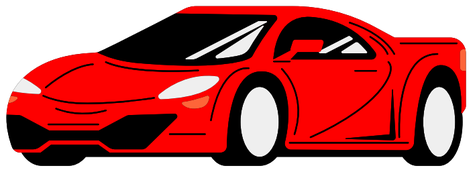
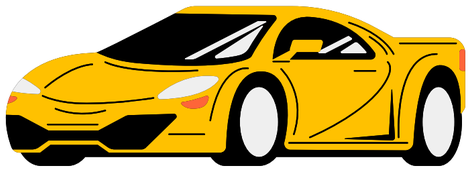
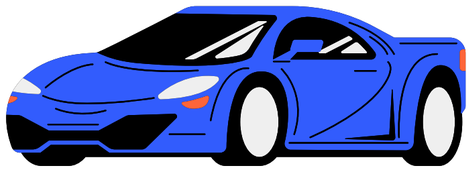
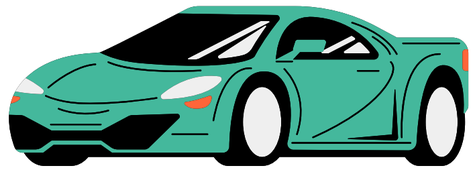
3. CONSTRUCTOR

4. `THIS`

START



OBJECT-ORIENTED PROGRAMMING

WHAT IS OBJECT-ORIENTED PROGAMMING ?

- A programming paradigm
- Based on the concept of "objects"
- Contain data (attributes) and methods (functions) that operate on the data
- Enables modular, reusable, and maintainable code

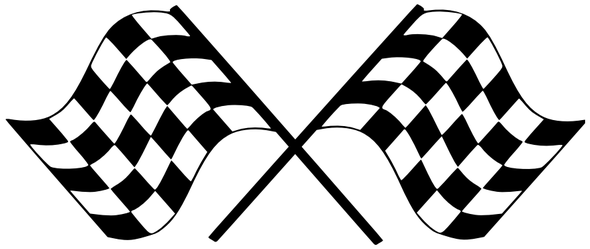
BENEFITS OF OOP

1. Modularity and Code Organization: Code is more organized and easier to manage

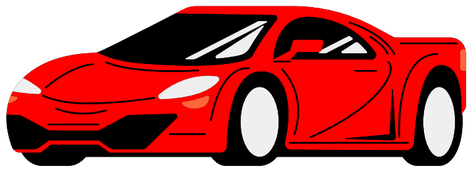
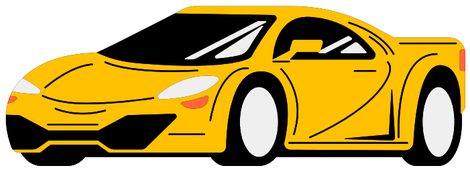
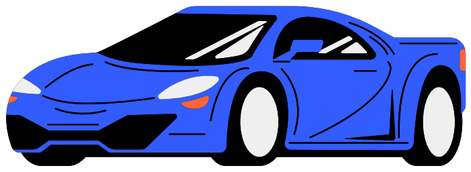
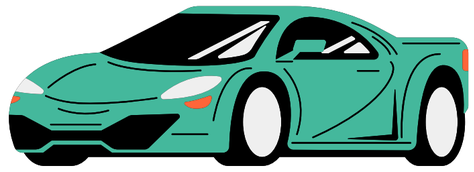
2. Code Reusability: Allows some parts of the code to be reused

3. Scalability: Development of large and complex systems are facilitated by creating code that is easily extendable

4. Enhanced Security: Use of access modifiers (e.g: private, protected) helps in enforcing strict control over how data is accessed and modified



CLASSES AND OBJECTS

WHAT ARE CLASSES AND OBJECTS ?

- Classes: Blueprint for objects. Defines the structure (attributes) and behavior (methods). Created with the `class` keyword

- Objects: Instances of classes. Created from a class

EXAMPLE OF CLASSES AND OBJECTS

```
class Car {
    String color;
    String model;

    void startEngine() {
        System.out.println("Engine started");
    }
}

public class Main {
    public static void main(String[] args) {
        Car myCar = new Car();
        myCar.color = "Red";
        myCar.model = "Tesla Model S";
        myCar.startEngine();
    }
}
```

Car is the class

color and model are the attributes

startEngine is the method

myCar is the object

EXAMPLE OF CLASSES AND OBJECTS

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```

Now, in order to instantiate an object, why does the class name have to be mentioned twice ?

EXAMPLE OF CLASSES AND OBJECTS

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    String color;
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        myCar.color = "Red";
        myCar.model = "Tesla Model S";
        myCar.startEngine();
    }
}
```

Now, in order to instantiate an object, why does the class name have to be mentioned twice ?

First occurence (Car myCar): Declare a variable named `myCar` that will hold a reference to an object of type `Car`

Second occurence (new Car()): Create a new instance (object) of the `Car` class

USING MULTIPLE CLASSES

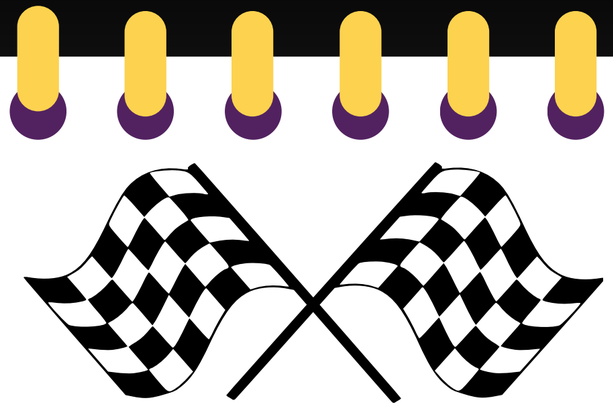
```
// Main.java

public class Main {
  int x = 5;
}
```

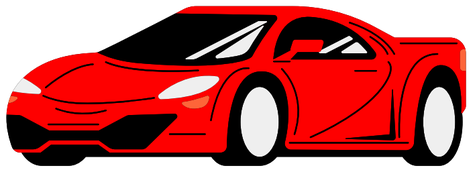
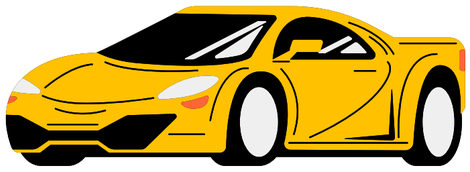
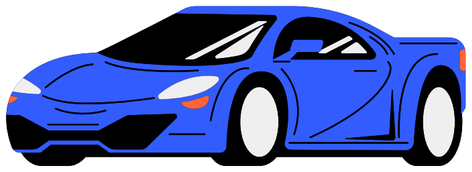
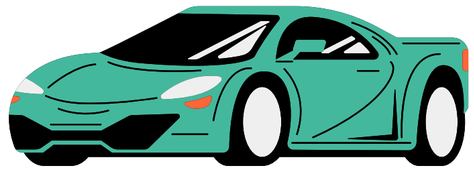
```
// Second.java

class Second {
  public static void main(String[] args) {
    Main myObj = new Main();
    System.out.println(myObj.x);
  }
}
```

Possible to create an object of class and access it in another class. Often used for better class organization



CONSTRUCTOR

WHAT IS CONSTRUCTOR ?

- Special method used to initialize objects
- Called when an instance (or object) of a class is created
- Sets up the initial state of the object, typically by assigning values to its fields (or attributes)

CHARACTERISTICS OF A CONSTRUCTOR

- Same Name as Class: A constructor must have the same name as the class in which it is defined
- No Return Type: Unlike regular methods, constructors don't have a return type, not even void
- Automatic Call: When an object is created using the new keyword, the constructor is called automatically
- Types of Constructors:
  - Default Constructor: A no-argument constructor that initializes fields with default values (e.g., null for objects, 0 for integers)
  - Parameterized Constructor: A constructor with parameters that allows initializing objects with specific values

CONSTRUCTORS

```
public class Book {
    String title;
    String author;

    public Book() {
        this.title = "Unknown";
        this.author = "Unknown";
    }

    public Book(String title, String author) {
        this.title = title;
        this.author = author;
    }

    public void displayBookDetails() {
        System.out.println("Title: " + this.title);
        System.out.println("Author: " + this.author);
    }
```

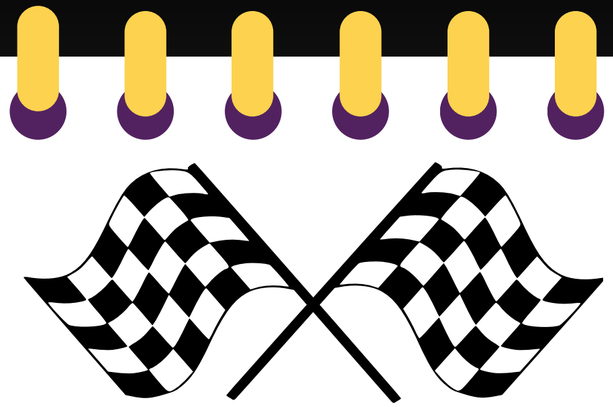
Default Constructor

Parameterized Constructor

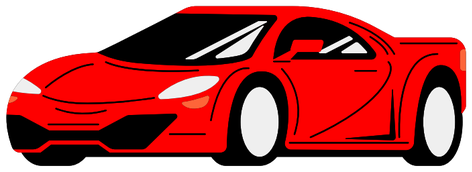
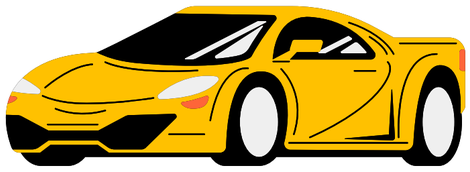
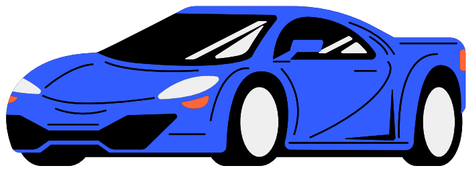
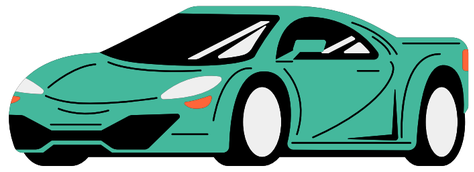
```
public static void main(String[] args) {
        Book book1 = new Book();
        Book book2 = new Book("Depressing Emotions", "Tom Cruise");
        book1.displayBookDetails();
        book2.displayBookDetails();
    }
}
```

book1 will output Unknown as no values are provided

book2 will output the set title and author



'THIS'

WHAT IS `THIS`?

- Used in Java to refer to the current instance of the class

- Uses:
  - Distinguishing Between Variables: When a parameter in a method or constructor has the same name as a class field, `this` helps to refer specifically to the field of the current object
  - Calling Another Constructor: `this()` can be used to call one constructor from another constructor within the same class, allowing for code reuse
  - Passing the Current Object: `this` can be used to pass the current object to other methods or constructors
  - Returning the Current Object: `this` can also be used to return the instance itself from a method

'THIS' IN CONSTRUCTOR

```
public class Book {
    String title;
    String author;

     public Book() {
       this.title = "Unknown";
       this.author = "Unknown";
    }

      public Book(String title, String author) {
       this.title = title;
       this.author = author;
    }

    public void displayBookDetails() {
        System.out.println("Title: " + this.title);
        System.out.println("Author: " + this.author);
    }
```

this.title and this.author refers to the title and author defined in the class

'THIS' TO PASS CURRENT OBJECT

```
class Person {
    private String name;

    public Person(String name) {
        this.name = name;
    }

    public void befriend(Person otherPerson) {
        System.out.println(this.name + " is now
friends with " + otherPerson.name);
    }

    public void makeFriendWith(Person
friend) {
        friend.befriend(this);
    }
}
```

```
public class PersonMain {
    public static void main(String[] args) {
        Person person1 = new Person("Alice");
        Person person2 = new Person("Bob");

        person1.makeFriendWith(person2);
    }
}
```

1. Method Call:

person1.makeFriendWith(person2);

- person1 is calling the makeFriendWith method, and it passes person2 as an argument

- Inside makeFriendWith, friend refers to person2

'THIS' TO PASS CURRENT OBJECT

```
class Person {
    private String name;

    public Person(String name) {
        this.name = name;
    }

    public void befriend(Person otherPerson) {
        System.out.println(this.name + " is now
friends with " + otherPerson.name);
    }

    public void makeFriendWith(Person
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        friend.befriend(this);
    }
}
```

```
public class PersonMain {
    public static void main(String[] args) {
        Person person1 = new Person("Alice");
        Person person2 = new Person("Bob");

        person1.makeFriendWith(person2);
    }
}
```

2. Inside makeFriendWith:
   - In this method, this refers to the object that called makeFriendWith. In this case, person1 called it, so this refers to person1
   - When friend.befriend(this); is executed:
     - friend refers to person2
     - this refers to person1
   - The line friend.befriend(this); translates to person2.befriend(person1)

'THIS' TO PASS CURRENT OBJECT

```
class Person {
    private String name;

    public Person(String name) {
        this.name = name;
    }

    public void befriend(Person otherPerson) {
        System.out.println(this.name + " is now
friends with " + otherPerson.name);
    }

    public void makeFriendWith(Person
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        friend.befriend(this);
    }
}
```

```
public class PersonMain {
    public static void main(String[] args) {
        Person person1 = new Person("Alice");
        Person person2 = new Person("Bob");

        person1.makeFriendWith(person2);
    }
}
```

3. Calling befriend:
   - person2.befriend(person1); is executed, originated from friend.befriend(this)
   - Inside befriend, this now refers to person2 (since befriend was called on person2), and otherPerson is person1

`THIS` TO PASS CURRENT OBJECT

```
class Person {
    private String name;

    public Person(String name) {
        this.name = name;
    }

    public void befriend(Person otherPerson) {
        System.out.println(this.name + " is now
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    }

    public void makeFriendWith(Person
friend) {
        friend.befriend(this);
    }
}
```

```
public class PersonMain {
    public static void main(String[] args) {
        Person person1 = new Person("Alice");
        Person person2 = new Person("Bob");

        person1.makeFriendWith(person2);
    }
}
```

4. Final Output:
   - Inside befriend, this.name is person2.name (which is "Bob")
   - otherPerson.name is person1.name (which is "Alice")

`THIS` TO RETURN CURRENT OBJECT

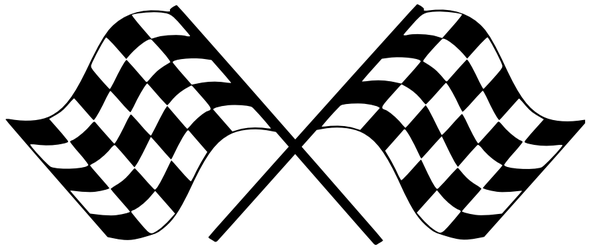
```
class Person {
    private String name;
    private int age;

    public Person(String name, int age) {
        this.name = name;
        this.age = age;
    }

    public String getDescription() {
        return name + " is " + age + " years old.";
    }

    public Person getCurrentPerson() {
        return this;
    }
}
```

```
public class Main {
    public static void main(String[] args) {
        Person person1 = new Person("Alice",
30);

        Person currentPerson =
person1.getCurrentPerson();

System.out.println(currentPerson.getDescri
ption());
    }
}
```

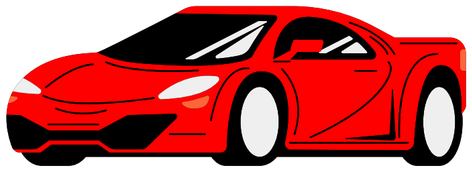
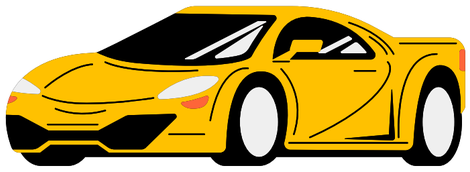
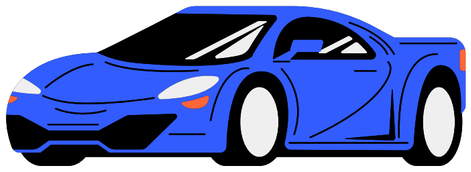
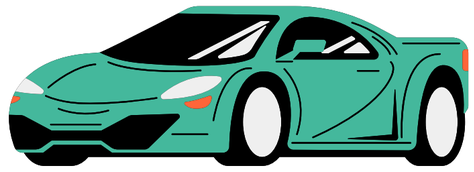
1. Creating the Person: person1 is created with the name "Alice" and age 30

2. Getting the Current Instance: When person1.getCurrentPerson() is called, it returns `this`, which refers to person1 itself. This instance is assigned to currentPerson

3. currentPerson.getDescription() will result in “Alice is 30 years old”



ANY QUESTION?

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